Paleomagnetism and age of three Canadian Rocky Mountain diatremes[1]

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Received March 3, 1991
Revision accepted August 9, 1991

Paleomagnetic results, and age estimates derived from them, are presented for three diatremes, using as a basis of comparison the combined apparent polar wander (APW) path for North America and Europe of Van der Voo. The Cross diatreme of the Front Ranges of the Canadian Rocky Mountains has yielded a radiometric age of 241 Ma (earliest Triassic) and is hosted by the flat-lying Pennsylvanian Tunnel Mountain Formation. It has normal polarity magnetization and yields a paleopole correctly placed according to its radiometric age on the APW path. The Blackfoot diatreme (for which no radiometric age is available), which is located in the Main Ranges of the Rocky Mountains, is known to be post-Late Ordovician because it is hosted by rocks of that age. It also has magnetization of normal polarity and yields a paleopole that, when calculated with respect to present horizontal, is coincident with the latest Cretaceous to Paleocene paleopole for North America. The paleopole, when calculated with respect to bedding, lies on the Middle Ordovician portion of the combined APW path. A clockwise rotation of 10° brings the paleopole into agreement with the latest Ordovician. Hence, from a paleomagnetic standpoint, a latest Cretaceous to Paleocene or latest Ordovician age is possible. The HP pipe (radiometric age 391 ± 5 Ma or Early Devonian), previously studied by D. T. A. Symons and M. T. Lewchuk, is hosted in limestones of Upper Cambrian to Middle Ordovician age. It has reversed polarity and yields a paleopole that, when compared with the combined APW path, suggests an age of mid-Permian, although errors are such that it could be somewhat younger, roughly coeval with the Cross diatreme. We conclude, therefore, that the radiometric age estimated for the HP pipe could be too old by about 130 million years.

Les résultats d'une étude paléomagnétique, ainsi que les âges qui en sont déduits, sont présentés pour trois diatrèmes, en utilisant comme repère la courbe de la dérive apparente des pôles de l'Amérique du Nord et de l'Europe de Van der Voo. Le diatrème Cross, localisé dans les chaînes Frontales des montagnes Rocheuses canadiennes, a fourni un âge radiométrique de 241 Ma (début du Trias), il recoupe les roches horizontales de la Formation de Tunnel Mountain, du Pennsylvanien. Son aimantation possède une polarité normale, il fournit un paléopôle correctement d'après l'âge radiométrique sur la courbe de la dérive apparente des pôles. Le diatrème Blackfoot (âge radiométrique inconnu), lequel apparaît dans les chaînes Principales des montagnes Rocheuses, est assigné post-Ordovicien tardif car il recoupe les roches de cet âge. Son aimantation est de polarité normale et il fournit un paléopôle qui, lorsque calculé par rapport à l'attitude horizontale actuelle, coïncide avec le paléopôle de la fin du Crétacé à Paléocène de l'Amérique du Nord. D'autre part, si ce paléopôle est calculé relativement au litage, il tombe sur le segment de l'Ordovicien moyen de la courbe de la dérive apparente des pôles combinée. Une rotation de 10° place le paléopôle en une position qui correspond à la fin de l'Ordovicien. Donc, du point de vue paléomagnétique, l'âge pourrait être la fin du Crétacé à Paléocène ou possiblement la fin de l'Ordovicien. La pipe HP (âge radiométrique de 391 ± 5 Ma ou Dévonien précoce), étudiée antérieurement par D. T. A. Symons et M. T. Lewchuk, recoupe les calcaires du Cambrien supérieur et de l'Ordovicien moyen. Sa polarité est inversée, et elle fournit un paléopôle qui, comparée avec la courbe de dérive apparente des pôles combinée, suggère comme âge le Permien moyen, quoique les erreurs sont telles que ce paléopôle pourrait bien correspondre à un âge un peu plus jeune, et grossièrement contemporain du diatrème Cross. En conclusion, l'âge radiométrique estimé pour la pipe HP serait trop vieux d'environ 130 millons d'années.

[Traduit par la rédaction]

Can. J. Earth Sci. **29**, 35–47 (1992)

Introduction

Three diatreme provinces have been recognized in the Canadian portion of the Foreland Fold and Thrust Belt (Pell 1986). From north to south they are the Williston Lake, Columbia Icefield, and Cranbrook–Invermere provinces. They are elongate parallel to the trend of the Rocky Mountains. We have studied the paleomagnetism of diatremes from the two southern provinces. All of them are small bodies.

Diatremes are notoriously difficult to date radiometrically. Radiometric age estimates are available from only 3 of the over 50 known occurrences in the two southern provinces: the HP pipe (Columbia Icefield province), and the Cross and the Blackfoot diatremes (Cranbrook–Invermere province). Estimated ages span 150 Ma. Consequently, many questions remain to be answered. For example, are the diatremes in the Columbia Icefield older than those in the Cranbrook–Invermere province? Were there repeated pulses of magmatism spread out over a long interval, or is the activity restricted in time? Is it possible that some diatremes were intruded during or after thrusting? Diatremes often are excellent recorders of the paleofield (Jones 1968; McFadden and Jones 1977; Symons and Lewchuk 1989) and it was our hope that paleomagnetism might help answer some of these questions about timing.

To carry out this study two conditions must be met. First it is necessary to define the reference paleofield for the Mesozoic and Paleozoic (apparent polar wander or APW path) with

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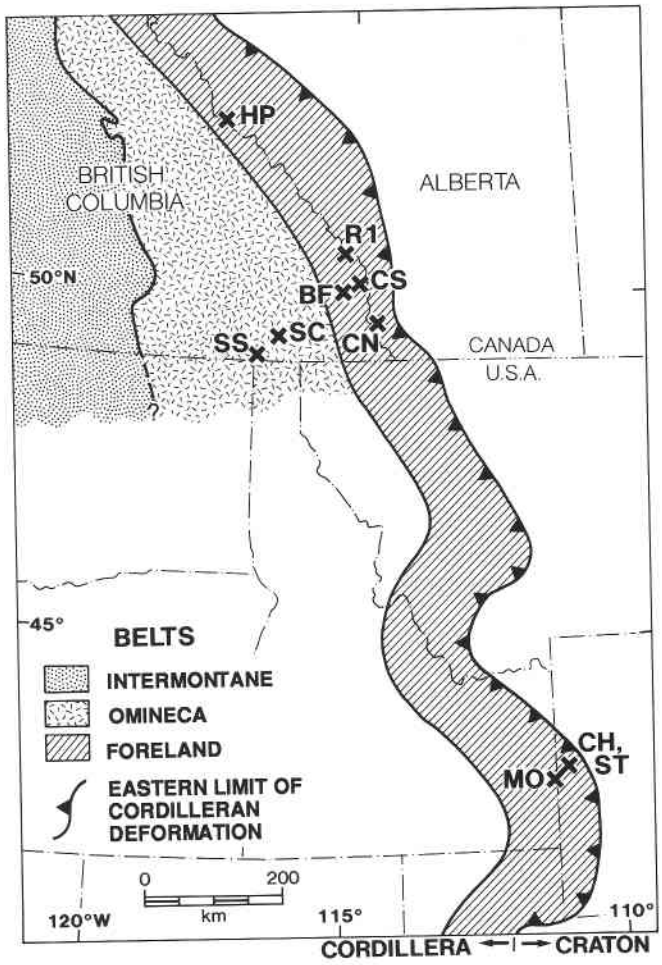


FIG. 1. Paleomagnetic sampling localities in the Phanerozoic rocks of the Foreland and eastern Omineca belts. Diatremes sampled are CS, Cross; HP, HP; BF, Blackfoot; R1, Rus 1. CN, Crowsnest Formation; SC, Skelly Creek Batholith; SS, Summit stock; CH, Chugwater Formation; MO, Lower Morrison Formation; ST, Stump Formation.

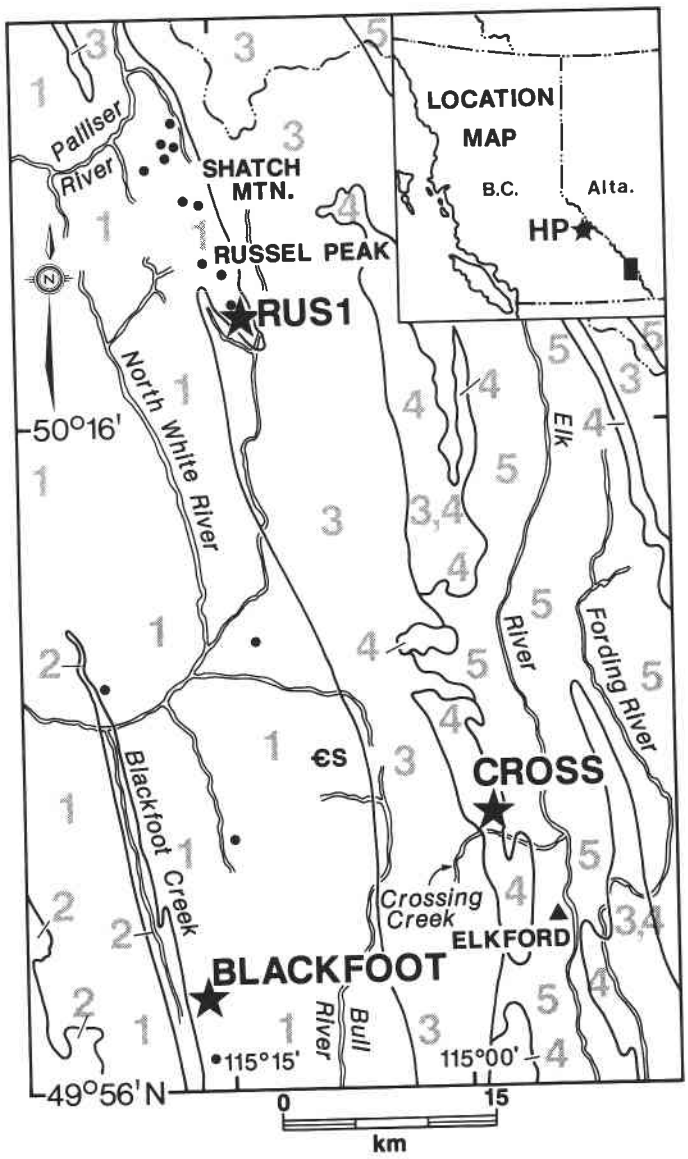


FIG. 2. Location of diatremes (stars) and simplified geology. Dots mark small diatremes and dikes not sampled. 1, Upper Cambrian to Silurian units; 2, Middle Devonian units; 3, Middle and (or) Upper Devonian to Mississippian units; 4, Permian and Pennsylvanian units; 5, Mesozoic units. Geology from Pell (1987) and Leech (1960, 1979). Diatreme and dike locations from Pell (1987).

which paleomagnetic data from the diatremes can be compared. A substantial part of this paper, therefore, is devoted to this task.

The second condition is that the Foreland Belt must not have undergone large north–south motions relative to the craton. There are very strong grounds for believing this to have been so. In the first place, geological ties to cratonic North America are well established (Price 1981). Also, four paleomagnetic studies from the Foreland Belt and the eastern Omineca Belt have yielded data generally concordant with the APW path for cratonic North America. About a decade ago, studies of Triassic red beds from the Rocky Mountains of Wyoming and Idaho, which are also part of the Foreland Belt (CH of Fig. 1), yielded data in excellent latitudinal agreement with those from cratonic North America (Grubbs and Van der Voo 1976). Results from Cretaceous and Jurassic rocks in the same region (MO, ST, Fig. 1) also were roughly concordant (Schwartz and Van der Voo 1984), but there is some possibility that they represent partial or complete overprints (Globerman and Irving 1988). More recently, a study by Symons and Lewchuk (1989) of the HP pipe (Fig. 1) and its host Cambro-Ordovician limestone has yielded data that they interpreted as being concordant with cratonic data and indicative of an Early Carboniferous age. Finally, Irving and Archibald (1990) obtained concordant data from two mid-Cretaceous intrusions from the eastern Omineca Belt (SC, SS, Fig. 1). The intrusions are well dated (Skelly Creek batholith, 99–94 Ma; Summit Stock, 102 Ma), and bathozonal evidence allows the paleodirections to be corrected for postemplacement tilting.

In contrast, results obtained from the Albian Crowsnest Formation (pyroclastic rocks from the Front Ranges) showed lower than expected inclinations and latitudes (Irving *et al.* 1986). These authors argued that the most probable explanation was not latitudinal displacement but rather that the unit represents only a single volcanic event, insufficient to provide an adequate time average of paleosecular variation.

Hence it seems probable, on paleomagnetic as well as geological grounds, that the second condition is met, and that the

TABLE 1. Mean paleopoles for North America and Europe combined, 30 Ma window

Mean age	N	Mean pole Lat. (°N)	Mean pole Long. (°E)	K	A_{95}	A_{63}
***(071)	5	79	190	326	4	2
160(159)	15	70	122	38	6	4
170(167)	10	70	111	52	7	4
180(190)	14	65	090	76	5	3
190(196)	18	65	088	90	4	2
200(202)	23	64	085	71	4	2
210(212)	19	58	087	37	6	3
220(220)	17	54	090	40	6	3
230(234)	29	51	106	63	3	2
240(242)	27	51	114	107	3	2
250(246)	28	50	114	95	3	2
260(269)	40	46	122	96	2	1
270(274)	43	45	124	84	2	1
280(278)	46	44	125	127	2	1
290(293)	28	41	128	187	2	1
300(299)	24	40	128	135	3	1
310(307)	20	38	130	128	3	2
320(322)	12	32	131	65	5	3
330(330)	11	30	132	60	6	3
340(340)	9	28	132	64	7	4
350(351)	7	29	130	80	7	4
360(367)	14	29	115	20	9	5
370(371)	15	28	112	27	8	4
380(376)	16	25	109	25	8	4
390(395)	16	14	108	22	8	5
400(403)	22	9	104	18	7	4
410(412)	33	13	110	15	7	4
420(419)	26	14	115	13	8	5
430(427)	21	19	127	18	8	4
440(441)	10	17	139	32	9	5
450(452)	12	18	146	28	8	5
460(457)	8	17	146	31	10	5
470(468)	7	17	155	34	11	6
480(487)	4	15	160	40	15	7
490(487)	4	15	160	40	15	7
500(505)	5	8	162	45	12	6
510(514)	3	4	164	41	20	8
520(519)	4	7	160	31	17	8
530(538)	2	17	144	233	17	5
540(548)	5	14	153	30	14	7
550(554)	5	12	160	19	18	9
560(564)	7	5	167	22	13	7
570(575)	7	11	176	6	27	14
580(582)	8	11	172	6	24	13
590(587)	5	20	173	4	42	21

NOTES: The age of the centre of the window is given and the mean of the assigned ages of N paleopoles is given in parentheses. The assigned age for each individual paleopole has been calculated as the arithmetic mean of the upper and lower limits given in Tables 1 and 2 of Van der Voo (1990). These lists contain a total of 383 paleopoles, 308 from stable cratonic areas with ages from Early Cambrian to Late Jurassic. The time scale of Palmer (1983) is used. Radiometric ages are used when cited. Not included in our analysis are those entries in Tables 1 and 2 of Van der Voo where his quality index Q is <3 (10 paleopoles excluded) or the entry is asterisked (34 paleopoles excluded which for some reason were not suitable to be included in an APW path, see Van der Voo for details); errors not given in originals (16 paleopoles excluded) or errors $>16°$ (16 paleopoles excluded). Also, the following paleopoles were deemed derived from duplicate studies of others included in Van der Voo's list and were not used for calculating running means: Chugwater Formation (Herrero-Brevera and Helsley 1983) and Exeter Lavas (Cornwell 1967). The age of the Duncansby Neck paleopole (Storetvedt *et al.* 1978*a*) is very uncertain and so it, too, was omitted from the running mean calculations. A list of those entries in which our estimate of assigned age differs from that of Van der Voo (1990) is given in the appendix. ***(071) is the mean latest Cretaceous to Paleocene paleopole for North America as documented in Table 3. N, number of paleopoles in window; K, their precision; A_{95}, radius of 95% confidence circle for mean paleopole; A_{63}, radius of 63% confidence circle for mean paleopole.

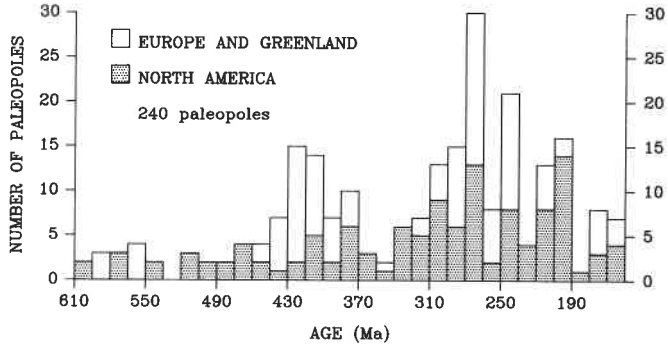


FIG. 3. Time distribution of paleopoles used in the running mean averages of Table 1.

eastern part of the Ominecas and the Rocky Mountains have not moved a great distance, in a north–south sense, relative to North America.

Sampling

Eight diatremes were sampled. Three of them, Cross (50°05′N, 114°58′W), Blackfoot (49°58′N, 115°16′W), and HP (51°41′N, 116°58′W), yielded acceptable data (Fig. 2). Rus 1 (50°25′N, 115°13′W) gave quasi-coherent data. The remainder did not yield coherent data. A total of 46 oriented hand samples were collected from the four diatremes, all of which are located at high altitude in rugged terrain: 10 from HP (Columbia Icefield diatreme province), 11 from Blackfoot, 10 from Cross, and 15 from Rus 1 (Cranbrook–Invermere province).

APW path

In a recent paper Van der Voo (1990) has shown, by comparing paleomagnetic poles (paleopoles) from Paleozoic to Early Jurassic rocks of Europe, Greenland, and North America, using a variety of continental reconstructions, that the assembly of Bullard *et al.* (1965) provides the best fit; on the basis of that reconstruction he showed that the North American and European APW paths agree exceedingly well (mean arc length difference between coeval paleopoles in a north–south sense is $+1.5°$, and that in an east–west sense is $-0.3°$, both statistically insignificant). This very important result means that a much improved APW path for cratonic North America can now be obtained by rotating all European paleopoles into the North American frame of reference using the rotation parameters of Bullard *et al.* (1965). These parameters are as follows: rotation of Greenland to Europe 22° about 73.0°N, 96.5°E; Europe to North America −38.1° about 88.4°N, 27.7°E. We refer to this common path as the "combined" APW path. Van der Voo computed such a path, which we use below. We also present alternative calculations to determine how the comparisons we make are influenced by the procedures adopted.

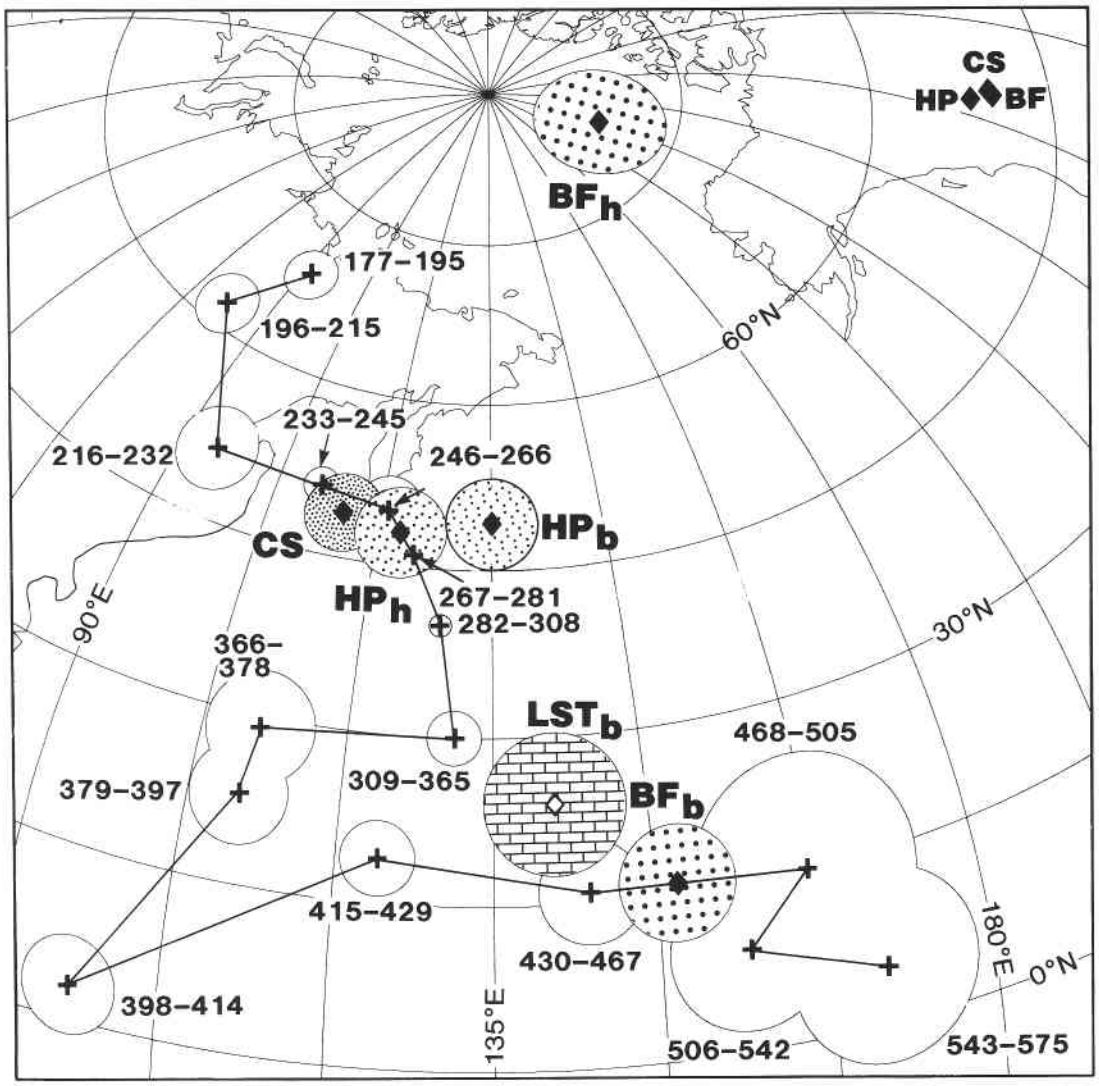


FIG. 4. Combined APW path of Van der Voo (1990, Table 7). Age ranges (Ma) are given. Standard error circles are drawn ($P = 0.37$). CS, Cross paleopole with respect to present horizontal (bedding is horizontal); BF_h and BF_b, Blackfoot paleopole with respect to present horizontal and tilt corrected respectively; HP_h and HP_b paleopole (revised from Symons and Lewchuk 1989) with respect to present horizontal and tilt corrected respectively; LST_b, paleopole of the host limestone of HP pipe, tilt corrected from Symons and Lewchuk (1989).

The data are not uniformly distributed in time but tend to be bunched at certain time intervals (Fig. 3). As a consequence, Van der Voo has obtained mean paleopoles for these intervals (Van der Voo 1990, Table 7). His mean paleopoles are plotted in Fig. 4. Alternatively, a running mean can be calculated using a moving window. This has been done for a 30 Ma window and increments of 10 Ma (Figs. 5 and 6, Table 1). Van der Voo (1990, Tables 1, 2) has provided an extensive list of data available and has made age assignments. We have reviewed the original papers and have, as a consequence, made minor revisions to his age assignments. The changes are listed in the Appendix. The arithmetic mean of the age range was used as the assigned age for each paleopole obtained by the running mean calculations. In compiling average paleopoles we have excluded some data given in Tables 1 and 2 of Van der Voo (1990). Our reasons for doing so are given in the footnote to Table 1.

The path of Van der Voo (Fig. 4) reflects the irregular temporal spread of data. The path obtained by a moving window also reflects this nonuniformity but in a different way: intervals with fewer data yielding larger errors. The nonuniformity in the age distribution of paleopoles is also expressed in Table 1 by the difference between the age of the centre of the window and the mean age of rocks studied. Both Figs. 4 and 5 display the same general characteristics: long curvilinear polar tracks punctuated by sharp bends, which have been variously called hairpins (Irving and Park 1972) or cusps (Gordon *et al.* 1984). Both procedures preserve the tracks, but averaging procedures of any sort will, of course, tend to smooth out hairpins. In the present context this is not important because comparisons made are with tracks. The general morphology of the two paths is similar (compare Figs. 4 and 5), and the purpose of giving both is to show that the interpretation of our diatreme data is not dependent on their method of calculation. In the present context our preference is for the running mean combined APW path because calculating a reference paleopole every 10 Ma allows a closer estimate to be made of the age of magnetization of diatremes.

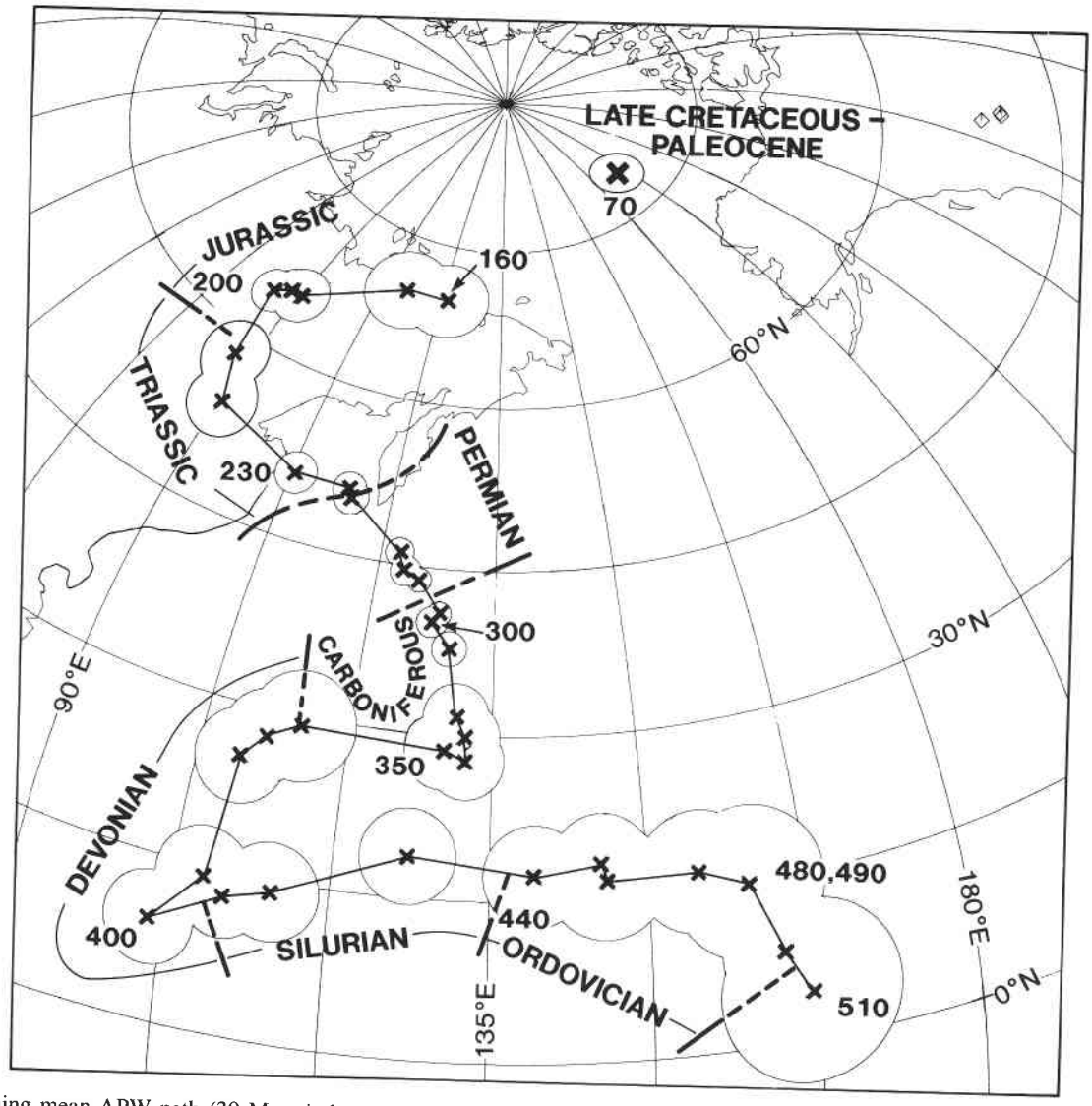


FIG. 5. Running mean APW path (30 Ma window, 10 Ma increments) as listed in Table 1. The latest Cretaceous to Paleocene paleopole (70 Ma) is documented in Table 3. Period limits are from Palmer (1983). Standard errors ($P = 0.37$) are plotted.

The Cross diatreme

This body intruded carbonates of the Pennsylvanian Tunnel Mountain Formation (McGugan and Rapson 1962). Three radiometric dates of 244 Ma (Grieve 1982), 241 ± 5 Ma (Smith *et al.* 1988), and 234 ± 5 Ma (Alibert and Albarede 1988)), all obtained using the Rb−Sr method on phlogopite, indicate an earliest Triassic age. The host rocks occur in the Bourgeau thrust sheet of the Front Ranges (Grieve 1982). They are flat lying, and no correction of the paleomagnetic directions for geological tilt is required. The body is exposed in a very steep face and is approximately 70 m wide. Marginal contacts are subvertical and dip away from the core of the exposure (Grieve 1982). The body consists of at least eight intrusive phases, including both breccias and massive hypabyssal kimberlite (Hall *et al.* 1989). Three hypabyssal units were sampled.

The Eastern Massive Phase has moderate coercivities (20−40 mT) and unblocking temperatures in the range 400−500°C. On orthogonal diagrams, the remanence decays linearly to the origin. Except for one hand sample, all magnetizations are single component, and hence all have been cleaned at 500°C.

The Red-Spotted and Western Massive phases have similar magnetic properties, with coercivities somewhat lower than those of the Eastern Massive Phase. Unblocking temperatures are in the range 500−550°C. Specimens from these phases yield good decay lines with alternating field (AF) demagnetization, and the principal magnetization, therefore, was obtained by line fitting within the range of 10−60 mT. A typical specimen is shown in Fig. 7, where RM is the direction of the magnetization removed in the interval 10−40 mT.

The mean directions of magnetization of hand samples are plotted in Fig. 8*a*. All are normally magnetized. The directions of the Red-Spotted and Western Massive phases (crosses and squares respectively) are statistically identical, but the direction of the Eastern Massive Phase (triangles) is different ($P = 0.05$). The latter could represent a somewhat different time interval, so the overall average, obtained as the mean of hand-sample directions (Table 2), is not a single spot reading of the field but an average over an appreciable interval.

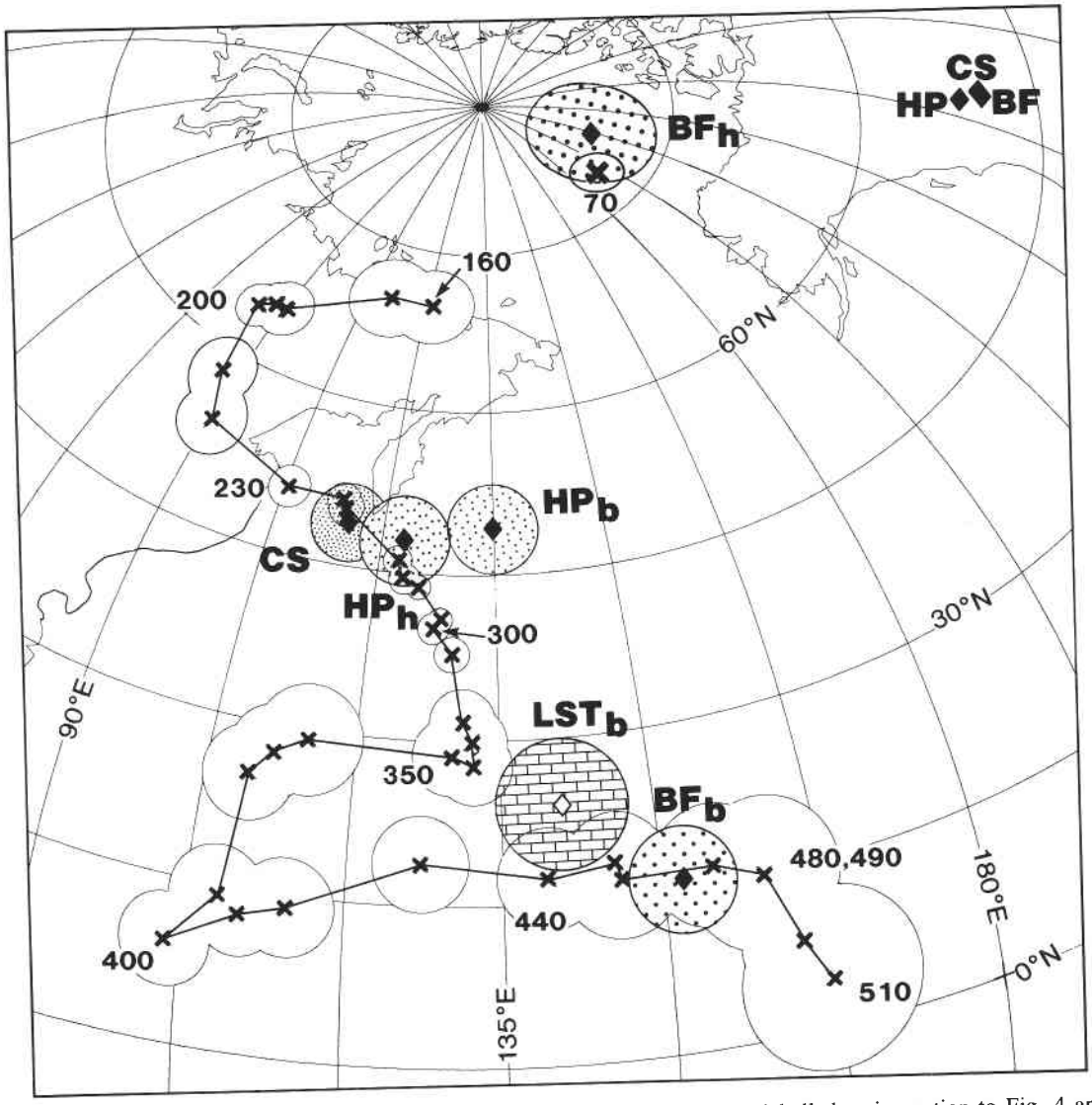


FIG. 6. Running mean APW path (as in Fig. 5) with the diatreme paleopoles added, labelled as in caption to Fig. 4 and Table 2.

The corresponding paleopole for the Cross diatreme (CS, Figs. 4, 6) falls on the Late Permian to Early Triassic segment of the combined APW path. Polarity is normal. The Permian geomagnetic field is consistently reversed, whereas frequent reversals and normal polarity begin to occur only in the latest Permian (Irving and Pullaiah 1976). Hence the polarity indicates a latest Permian or very earliest Triassic age. This is in excellent agreement with geological field relationships and with the radiometric ages of 241–234 Ma. The Permian–Triassic boundary is given as 245 Ma by Palmer (1983). These data indicate that there was no significant latitudinal displacement relative to the craton using either an earliest Triassic (240 Ma, relative paleolatitudinal displacement (RPD) = 1 ± 7°) or latest Permian (250 Ma, RPD = 0 ± 7°) combined reference pole. Similarly, no appreciable rotation relative to the craton is determined from these two reference poles (relative rotation (RR) = 2 ± 7° and 2 ± 11° respectively).

The Blackfoot diatreme

This body intrudes the uppermost Ordovician to lowermost Silurian Beaverfoot Formation (Ashgillian to Llandoverian), which dips 68° at an azimuth of 263°. The diatreme occurs in the Bull River thrust sheet, as defined by Price (1981), which is situated in the Main Ranges of the Rocky Mountains. The contact with overlying sedimentary rocks is not exposed, as the top of the diatreme lies at the present erosional surface. The surface exposure is shaped like an inverted "b." It is 400 × 400 m at the north end and tapers to a point 1000 m to the south (Pell 1986). Contact and contour intersections (taken from the map in Pell 1986) show that the eastern margin dips west at about 60° and the western margin dips east at about 40°, giving the intrusion a wineglass shape in cross section. These relationships may be interpreted to indicate that emplacement of the Blackfoot diatreme occurred after tilting of the host rocks. Zircons have yielded concordant U–Pb ages that are much too old (530 ± 2 Ma) and clearly reflect their inherited nature (Reichenbach and Parrish 1988). Hence, the age is poorly constrained as post-latest Ordovician to earliest Silurian (postdeposition of the Beaverfoot Formation).

The intrusion is a strongly foliated, severely altered body not amenable to petrographic classification, but the presence of aluminium-rich augite phenocrysts and pyroxenite xenoliths suggests an affinity with alkalic basalt. It consists of an intensely foliated phase intruded by thin breccia dikes and a

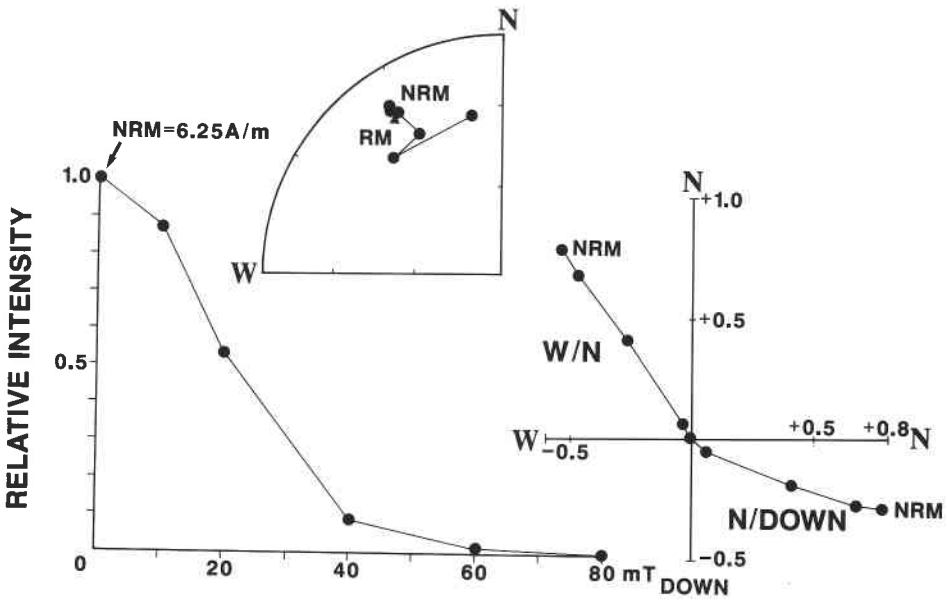


FIG. 7. AF demagnetization of the Red-Spotted Phase of the Cross diatreme with respect to present horizontal (beds are flat lying). Decay in intensity left, changes in direction centre, and orthogonal plots right. Solid dots on stereonet denote downward directions (positive inclinations). The solid triangle (RM) is the direction of magnetization removed between 10 and 40 mT.

moderately foliated breccia phase. The age and origin of the foliation is problematic. The dikes, breccia, and a sandstone autolith were sampled.

The magnetizations are essentially single component. The moderately foliated breccia phase and the thin breccia dikes have moderate coercivities, and unblocking temperatures distributed between room temperature and 550°C (Fig. 9). Above 100°C, the decay lines on orthogonal plots are straight and pass through, or very close to, the origin. Principal magnetizations of the breccias were obtained by line fitting in the range 300−500°C. A large (2 m) sandstone autolith yielded straight decay lines (after AF demagnetization) passing close to, but not through, the origin, and principal magnetizations (barred symbols in Fig. 8*b*) were obtained by line fitting in the range 20−60 mT. They are not significantly different from those of the breccia phases (Fig. 8*b*). The agreement of the directions of the autolith and dike phases with the moderately foliated breccia phase indicates that any anisotropy of magnetic susceptibility produced by the foliation has not skewed the remanence direction.

Directions and paleopoles calculated with respect to present horizontal and to the plane of bedding in host sedimentary rocks are plotted in Figs. 8 and 6. They represent two possible interpretations of the data, which are now described. The paleopole with respect to present horizontal (BF_h) (no bedding attitude correction) is not significantly different from a latest Cretaceous to Paleocene North American cratonic reference paleopole (see Table 3).

This suggests that the Blackfoot diatreme is latest Cretaceous to Paleocene in age. According to Mack and Jerzykiewicz (1989) there were three thrusting events during the latest Cretaceous and earliest Tertiary (the Laramide orogeny), and it was during these thrusting events (although which event is unknown) that the host rocks of the Blackfoot became tilted, prior to the intrusion of the Blackfoot.

When corrected for the attitude of host rocks the marginal contacts become vertical (west margin) and horizontal (east margin)—a configuration that might be considered unlikely for a diatreme. The corresponding paleopole lies on the Middle Ordovician portion of the combined APW path at about 465 Ma (BF_b, Fig. 6). This would implies that the diatreme is older than the Beaverfoot Formation, which it intrudes. If, on the other hand, the Blackfoot diatreme is assumed to be latest Ordovician (the age of the oldest Beaverfoot Formation) and compared with the 440 Ma paleopole (Table 1), then a small postemplacement counterclockwise rotation (RR = 10 ± 7°) with no significant latitudinal displacement is inferred.

The paleomagnetic data therefore are consistent with a latest Cretaceous to Paleocene age or a latest Ordovician age. If the former is correct, then the Blackfoot is the first post- or syn-Laramide igneous body to be recognized in the Main Ranges of the Rocky Mountains, and it becomes a prime target for radiometric work. Radiometric dating of the Blackfoot could help to document the upper age limit of deformation in that area.

The HP pipe

Pell (1986, 1987) mapped and described two principle lithologies in the HP pipe: a breccia phase (composed of five varieties of breccia) and a subordinate dike phase. The pipe intrudes Upper Cambrian to Middle Ordovician limestone (Price 1967), which dips variably to the south (0−20°, Symons and Lewchuk 1989; 5−30°, Pell 1986). The pipe contacts are almost vertical, as are the internal dike contacts (Symons and Lewchuk 1989). J. Harakal and K. Scott obtained two K−Ar dates from biotite of 391 ± 12 and 396 ± 10 Ma implying an Early Devonian age (R. L. Armstrong, personal communication, 1990). Pell (1987) reported a Rb−Sr date on phlogopite of 348 ± 7 Ma, implying an Early Carboniferous age. The latter, a preliminary result, has since been superseded by a Rb−Sr date of 391 ± 5 Ma or Early Devonian (R. L. Armstrong, personal communication, 1990), which agrees well with the earlier K−Ar results. Symons and Lewchuk's paper appeared before the second, and presumably more reliable, Rb−Sr date became available. In their paper,

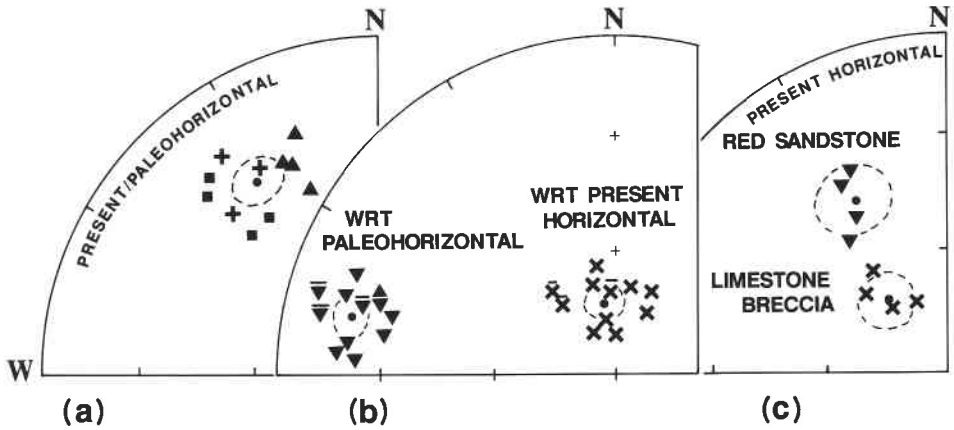


FIG. 8. Mean directions of hand samples, and overall mean directions with associated error circle (dashed) at $P = 0.05$ for (*a*) Cross diatreme, (*b*) Blackfoot diatreme, and (*c*) Rus 1 diatreme. In *a*, triangles denote data from Eastern Massive Phase; crosses from Red-Spotted Phase; and squares, from Western Massive Phase. In *b*, bar over symbol denotes data from a sandstone autolith, the remainder are from breccia phases.

TABLE 2. Mean directions

Diatreme	H	S	M ($\times 10^{-2}$ A/m)	D (°)	I (°)	k	α_{95} (°)	Lat. (°N)	Long. (°E)	dm (°)	dp (°)	A_{95} (°)
Cross (h = b)	11	78	13.3	328	33	52	6	49	115	7	4	6
Blackfoot (h)	11	60	0.57	351	72	78	5	81	211	9	8	9
Blackfoot (b)				283	20			16	152			
Rus1												
Sandstone (h)	4	22	0.52	329	41	97	9	54	117	11	7	9
Sandstone (b)				318	75			65	197			
Rus 1												
Breccia (h)	5	34	1.00	320	67	114	7	65	170	12	10	11
Breccia (b)				270	53			25	178			
HP												
S & L	26	75	—	136	−37	20	7	44	128	8	5	6
S & L + this study (h)	20	59	—	141	−39	25	7	48	123	8	5	7
S & L + this study (b)	—	—	—	135	−46	—	—	49	135	—	—	—
LST, limestone, S & L	7	20	—	114	−23	24	12	24	141	7	13	11

NOTES: H is number of hand samples, given unit weight in the averages; S is number of specimens; M is the mean magnetic intensity; D and I are the declination and inclination; k is best estimate of Fisher's precision parameter; α_{95} is radius of circle of confidence limit $P = 0.05$; lat. and long. are paleopole latitude and longitude; dm and dp are the semiaxes of the error ellipse; and A_{95} is the circular error of the paleopoles at $P = 0.05$. (h) and (b) indicate the directions and corresponding paleopoles are with respect to present horizontal and corrected for the tilt of host sediments respectively. Bedding attitudes (dip, downdip azimuth) used for the restoration to paleohorizontal are as follows: Blackfoot (68°, 263°); Cross (0°, 0°); Rus 1 red sandstone (30°, 230° host carbonate bedding followed by 37°, 116° rotated sandstone bedding); Rus 1 limestone breccia (30°, 230°), HP and LST (10°, 180°). S & L, data of Symons and Lewchuk (1989), *A* component magnetizations from dike and breccia samples. S & L + this study, hand samples with $\alpha_{95} < 20°$, 18 from Symons and Lewchuk (1989) and 2 collected in this study. The two hand samples collected in this study gave the following results: $S = 4$, $D = 155°$, $I = -42°$, $k = 59$, $\alpha_{95} = 12°$; $S = 4$, $D = 154°$, $I = -32°$, $k = 26$, $\alpha_{95} = 18°$).

based on a study more extensive than ours, they favoured an Early Carboniferous age, arguing that the K−Ar ages were too old and reflect excess initial argon. They identified three magnetizations: a pretilting magnetization (*A*) in the pipe breccia and dike rocks, which they interpreted as being primary; a pretilt magnetization (*B*) in the limestones, which they also interpreted as being primary; and a postintrusive overprint (*C*). The paleopole for the *B* magnetization (tilt corrected) touches the Late Ordovician of the combined APW path (LST_b, Fig. 6), a result that is consistent with the limestone's Upper Cambrian to Middle Ordovician age (Price 1967).

We have recalculated the mean direction of the *A* magnetization using only those hand samples (18 samples) from Symons and Lewchuk (1989) with errors (α_{95}) less than 20° and 2 samples from our collection. The mean direction is little changed (Table 2). The recalculated HP paleopole, with respect to present horizontal, is plotted in Fig. 6 (HP_h). It lies close to the Cross diatreme paleopole in the mid-Permian section of the combined APW path. Assuming that the more recent radiometric work is correct, and that the age of the HP diatreme truly is Early Devonian, then the HP paleopole is very different from the combined reference paleopole for the Early Devonian as given by Van der Voo (Fig. 4, labelled 398−414 Ma), and by us (Fig. 5, labelled 400 Ma). With respect to present horizontal, the recalculated HP result (HP_h) yields a RPD of $-52 \pm 10°$ and RR of $17 \pm 11°$ when compared with the 400 Ma combined paleopole (Table 1). When corrected for tilt of the host limestone's bedding (10° at 180°, Symons and Lewchuk 1989), the HP paleopole moves away from the APW path, and the displacement becomes even greater (HP_b, Fig. 6). This implies that the HP pipe has been displaced over 5000 km to the south, relative to the craton, if

TABLE 3. Latest Cretaceous and Paleocene cratonic reference paleopoles from North America

Rock unit, location, reference	Age (Ma)	Pole (°N)	Pole (°E)	A_{95} (°)
Alkalic Intrusions, Montana (N = 35), 1	64.3±2.1	81.8	181.4	5.4
Gringo Gulch Volcanics, Arizona, 2	63.3±1.8	77.0	201.0	1.5
Roskruge Volcanics, Arizona, 3	72.7±4.2	73.6	176.0	7.7
Adel Mountain Volcanics, Montana, 4	76.2±3.1	82.2	209.9	6.8
Elkhorn Volcanics, Montana, 5	80.2±2.4	80.3	189.5	9.6
Mean (N = 5)	71.3±2.7	79.2	189.9	4.2

NOTES: References: 1, Diehl *et al.* (1983); 2, 3, Vugteveen *et al.* (1981); 4, Gunderson and Sheriff (1991); 5, Diehl (1991). K for the mean is 326.

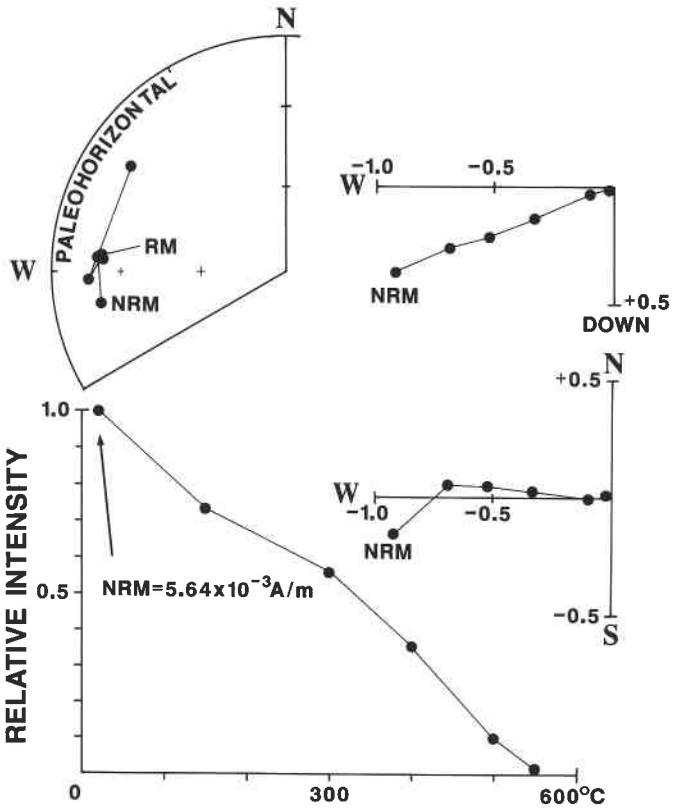


FIG. 9. Thermal demagnetization of breccia phase of the Blackfoot diatreme. Changes in directions above, intensity below, and orthogonal plots right. Solid dots on stereonet denote downward directions (positive inclinations). RM is the direction of the magnetization removed between 300 and 500°C.

it was in fact emplaced in the Early Devonian.

There are three reasons why a displacement of this magnitude is very unlikely. First, as already noted, the geological ties to the craton are well established (Price 1981). Second, the host carbonate yields data concordant with the craton (the *B* magnetization of Symons and Lewchuk (1989), plotted as LST_b in Fig. 6). Third, to the west of the Rocky Mountains in the eastern Omineca Belt the mid-Cretaceous Summit stock and Skelly Creek batholith have been shown to have magnetizations concordant with those of the craton, indicating that large-scale post-mid-Cretaceous relative latitudinal displacement is very unlikely.

What, therefore, is the reason for this very large discordance between the HP_h paleopole and the early Devonian combined reference paleopole? First, the paleomagnetic data could have been wrongly obtained. This is unlikely because our results agree well with those of Symons and Lewchuk (1989). Second, the early Devonian reference field could be grossly wrong. This is unlikely because it is based on numerous paleopoles determined by many laboratories from both North America and Europe (see the entries for 390 and 400 Ma in Table 1). Moreover, the discordance is so large as to be unaffected by the procedures used to calculate the reference paleopoles (compare Figs. 4 and 5). Third, the HP body could be Early Devonian in age and have been magnetically overprinted in the Permian. There is no evidence in the data that the *A* magnetization of Symons and Lewchuk (1989) is an overprint. A fourth possibility is that the radiometric age estimates are incorrect and the pipe is not Early Devonian but much younger (Fig. 6).

The changes in paleolatitude of North America, with time calculated using both the running mean and the Van der Voo (1990) combined APW paths, are plotted in Fig. 10. It is clear that based on paleomagnetic inclination alone the age of the HP diatreme could range from Early Jurassic (200 Ma) to Late Carboniferous (300 Ma), but could not be as old as Devonian. A mid-Permian age is most likely if rotations about a vertical axis are minimized (Fig. 6).

Assuming that the magnetization of the HP diatreme is mid-Permian, we propose the following sequence of events: (1) the Middle Ordovician limestone was deposited and magnetized, (2) the limestone was tilted, (3) HP was intruded in the Permian, and (4) thrust sheets were emplaced later with negligible tilting or rotation. This interpretation, that the diatreme was emplaced after the limestones were tilted, is supported by the near-vertical nature of diatreme and dike contacts. Also, there is evidence for pre-Permian tilt in the Main Ranges to the south where Mott *et al.* (1986) have identified an angular unconformity between the McKay Group (Late Cambrian to Early Ordovician) and the Beaverfoot Formation (latest Ordovician to earliest Silurian).

The Rus 1 diatreme

The Rus 1 diatreme (Pell 1987), located in the crest of an anticline of Beaverfoot Formation, is a funnel-shaped intrusion dominated by a breccia containing limestone clasts in a red siltstone matrix. It is overlain by a well-bedded red siltstone. A small outcrop of fresh basalt apparently occurs,

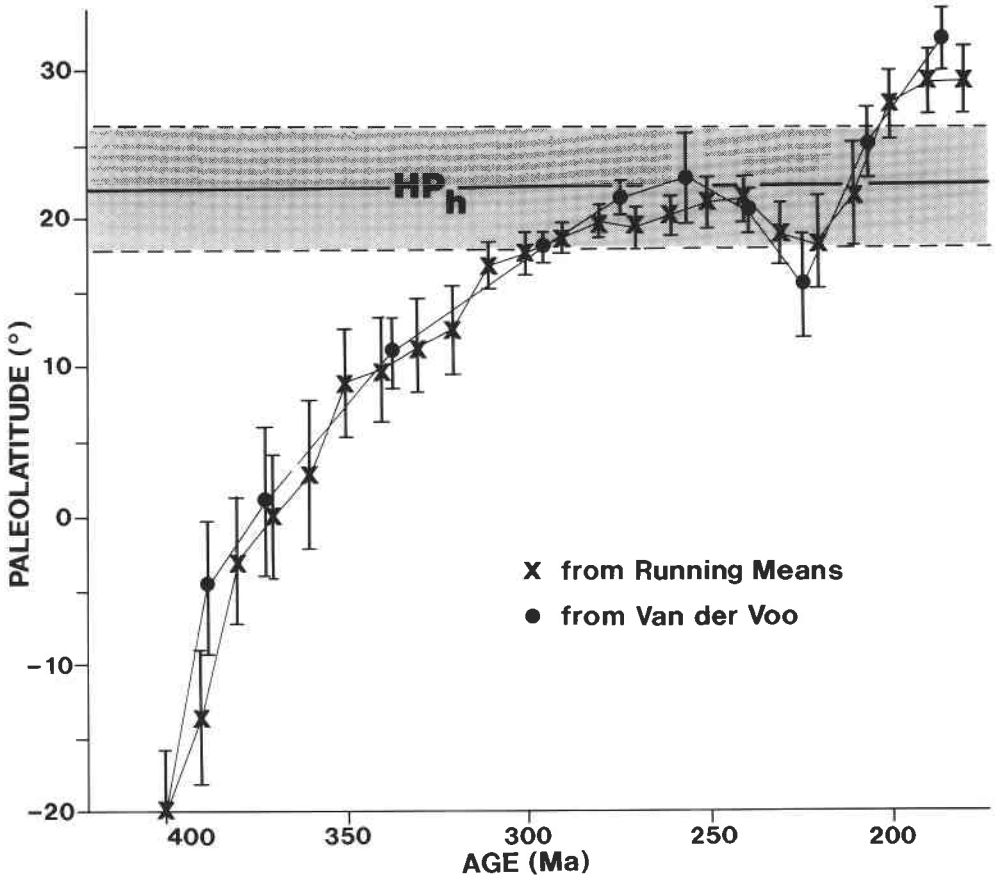


FIG. 10. Paleolatitude of HP pipe (standard errors $P = 0.37$) compared with North America (51°41′N, 116°58′W), with time calculated using the combined reference paleopoles listed in Table 7 of Van der Voo (1990) and the running mean averages of our Table 1. HP_h , the HP pipe paleolatitude calculated from the penultimate line of Table 2 (S & L + this study (b)).

above the siltstone, although the contact relationships are not clear. The diatreme is considered to have breached the pre-Devonian erosion surface (Helmstaedt *et al.* 1988). The siltstone is interpreted as a crater-fill deposit and, hence should be essentially contemporaneous with the breccia.

The basalt, collected from a rock pinnacle, is very fresh and contains unaltered olivines. It has a strong magnetization with random directions interpreted as having been caused by lightning. The limestone breccia has high coercivities (more than 30% of magnetization left after 100 mT) and unblocking temperatures between 300 and 650°C. The red siltstone has similar properties, but the directions are different from those of the breccia (Fig. 8). The cause of this discrepancy is not known and the data are not further interpreted.

Conclusions

(1) Radiometric and paleomagnetic data indicate an earliest Triassic age of emplacement for the Cross diatreme. (2) Paleomagnetic data indicate that the Blackfoot diatreme could be either latest Cretaceous to Paleocene or latest Ordovician in age. (3) Radiometric data from the HP pipe indicate an Early Devonian age, whereas paleomagnetic data indicate a Permian age. (4) Paleopoles of the Cross and HP diatremes are not statistically different ($P = 0.37$ error circles overlap, Fig. 6) and may therefore be of comparable (Permian–Triassic) age.

Acknowledgments

We would like to thank the reviewers of this paper: D. T. A. Symons, R. F. Butler, and C. J. Yorath. PJW thanks Jennifer Pell for her helpful introduction to the geology of the Rocky Mountain diatremes. DJS and HHH gratefully acknowledge grants from the Natural Sciences and Engineering Research Council of Canada. DJS, HHH, and DCH thank the Department of Energy, Mines and Resources Canada for support, through a research agreement.

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Appendix

In order to calculate his combined APW path Van der Voo (1990) assigned age limits and magnetization ages to all paleopoles. In the majority of instances we agree with his assignment. However, in the instances listed in Table A1 (36 of 240 paleopoles) we have modified his age assignments. The changes are small but we feel are worth recording because they represent an independent assessment of the data.

TABLE A1. Revised ages of paleopoles

Rock unit, location, reference	Range (Ma) Low	Range (Ma) High	Assigned age (Ma)	Δ
Summerville sandstone, Utah, Steiner (1978)			163	3
Watchung basalts, New Jersey – Pennsylvania, McIntosh *et al.* (1985)	198	208	203	2
New Jersey volcanics, New Jersey, Opdyke (1961)	198	208	203	2
Kayenta Fm., S. Utah, Johnson and Nairn (1972)	193	198	196	2
Kayenta Fm., N. Utah, Steiner and Helsley (1974)	193	198	196	2
Upper Maroon Fm., Colorado, McMahon and Strangway (1968)	240	296	268	2
Mauch Chunk synfolding, Pennsylvania, Kent and Opdyke (1985)	258	315	287	2
Dunkard Fm., Virginia, Helsley (1965)			286	2, 3
Prince Edward Island red beds, Roy (1966)	286	308	297	2
Prince Edward Island red beds, Black (1964)	286	308	297	2

TABLE A1 (*concluded*)

Rock unit, location, reference	Range (Ma) Low	Range (Ma) High	Assigned age (Ma)	Δ
Morien Group, Nova Scotia, Scotese *et al.* (1984)			286	2, 3
Traveler felsite, Maine, Spariosu and Kent (1983)	387	408	397	2
Eastport Fm., Maine, Kent and Opdyke (1980)			412	2
Hersey Fm., Maine, Kent and Opdyke (1980)			397	2, 3
Peel Sound Fm., Northwest Territories, Dankers (1982)	398	408	403	2
Malm gamma limestone, Germany, Heller (1978)	144	156	149	1
Pliensbachian sediments, United Kingdom, Hijab and Tarling (1982)	193	198	196	1
Liassic volcanics, France, Girdler (1968)	204	208	206	2
Brumunddal lavas, Norway, Storetvedt *et al.* (1978b)			245	3
Upper Lodève sandstones, France, Kruseman (1962)	258	265	262	2
Lodève Saxonian, France, Merabet and Guillaume (1988)	258	265	262	2
Lodève Autunian, France, Merabet and Guillaume (1988)	265	280	273	1, 2
Lower Lodève sandstones, France, Kruseman (1962)	265	280	273	1, 2
Moissey volcanics, France, Thompson *et al.* (1986)	258	280	269	1, 2
Arendal diabase, Norway, Halvorsen (1972)			286	3
Skaane quartz dolerites, Sweden, Bylund (1974)	286	333	310	1
Portishead red beds, United Kingdom, Morris *et al.* (1973)			360	3
Orkney lavas, United Kingdom, Storetvedt and Petersen (1972)			374	3
Esha Ness ignimbrite, United Kingdom, Storetvedt and Torsvik (1985)			374	3
Holy Cross Mountain sandstone, United Kingdom, Storetvedt and Torsvik (1983)	387	394	391	1, 2
Sarclet sandstone, United Kingdom, Storhaug and Storetvedt (1985)			387	3
Salrock Fm., Ireland, Smethurst and Briden (1988)	408	428	418	1
N. Highland zone 5, United Kingdom, Esang and Piper (1984)		<400	399	2
Strontian granite, United Kingdom, Torsvik (1984)	400	438	419	2
Fleming Fjord Fm., Greenland, Reeve *et al.* (1974)	208	230	219	2
Kap Stanton Fm., Greenland, Stearns *et al.* (1989)			408	3

NOTES: Range, best estimate of the lower and upper age limits of the paleopole. Assigned age is age used in the running mean. Assignments were changed for three reasons (last column): 1, to bring the age range into agreement with those of Palmer (1983); 2, the original paper suggests a different age range; 3, where the age is given as straddling a period boundary Palmer's boundary age is used, i.e., Permo-Triassic rocks are assigned an age of 245 Ma.